

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Artificial Intelligence and Data Science**  
**ASSESSMENT TOOL 3 – CODING CHALLENGE**

|                  |                                                                                                                                                          |               |                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------|
| Course Code:     | DSA0306                                                                                                                                                  | Course Title: | Natural Language Processing          |
| CO Assessed:     | CO1 – Imitation (L1)                                                                                                                                     | Assessment:   | Assessment Tool 3 – Coding Challenge |
| Weightage:       | 30% of CIA                                                                                                                                               |               |                                      |
| CO5 Statement:   | Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3) |               |                                      |
| Student Name:    | N.Tharun Sai                                                                                                                                             | Reg. No.:     | 192472382                            |
| Section / Batch: | 2024 slot c                                                                                                                                              | Date:         | 27-07-2026                           |

**Code of Conduct**

*I N.Tharun Sai (192472382) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: N.Tharun Sai

**Instructions**

- Develop a complete and modular Python program that meets all the functional requirements specified in the coding challenge using appropriate algorithms, data structures, and programming practices.
- Test the program using multiple valid and invalid test cases, and clearly display the input, processing, and output to verify the correctness of the implementation.
- Follow standard coding practices by using meaningful variable names, proper indentation, comments, exception handling (where applicable), and upload the source code, sample outputs, and README file to the designated GitHub repository.
- Duration: 40 minutes.

| Section / Topic | Focus                                                         | Q Nos |
|-----------------|---------------------------------------------------------------|-------|
| Porter Stemmer  | class design, methods, encapsulation, and rule-based stemming | 1     |

## Annexure A – Coding Challenge

### Section 1: Object-Oriented Implementation of a Porter Stemmer

You have recently joined an **NLP startup** called **LexiMind Technologies**. The company is developing a **search engine** for a digital library containing millions of research papers. Users expect the search engine to return relevant results even when different grammatical forms of a word are used.

For example, a search for **connect** should also retrieve documents containing **connected**, **connecting**, **connection**, and **connections**. To achieve this, the development team has decided to implement the **Porter Stemming Algorithm** from scratch.

Since the algorithm consists of multiple sequential rules (Step 1a, Step 1b, Step 1c, Step 2, Step 3, Step 4, Step 5a, and Step 5b), the project manager has divided the work among team members. Each developer is responsible for implementing and testing one specific rule. At the end of the project, all modules must be integrated into a single, fully functional stemmer.

#### Assessment Task

##### Phase 1: Individual Task (Assigned Rule)

Each student will be assigned **one Porter Stemmer rule**.

Your responsibilities are to:

1. Study and understand the assigned rule.
2. Explain the linguistic purpose of the rule.
3. Describe the conditions under which the rule should be applied.
4. Implement the rule in Python.
5. Prepare at least **10 test words** demonstrating:
  - Words that satisfy the rule.
  - Words that should remain unchanged.
6. Document the input, expected output, and actual output.
7. Explain any limitations or special cases encountered.

#### Rubrics for Calculation

| Criteria                       | Weightage | Level 4 – Exemplary                                                                       | Level 3 – Proficient                                      | Level 2 – Developing                          | Level 1 – Beginning                              |
|--------------------------------|-----------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------|--------------------------------------------------|
| <b>Problem Understanding</b>   | 20%       | Clearly understands problem constraints, edge cases, and competitive requirements         | Understands problem with minor gaps in constraints        | Partial understanding; misses key constraints | Misinterprets problem or constraints             |
| <b>Algorithm</b>               | 25%       | Selects optimal algorithm with best time and space complexity for competitive constraints | Uses efficient algorithm with minor scope for improvement | Uses basic/inefficient approach               | No clear algorithmic approach                    |
| <b>Code</b>                    | 20%       | Writes correct, efficient, and clean code that passes all constraints                     | Code works with minor errors or inefficiencies            | Partially correct code; fails for some cases  | Code does not execute or gives incorrect results |
| <b>Competitive Performance</b> | 20%       | Solves problem within time limits and handles large inputs effectively                    | Solves within acceptable time with minor delays           | Struggles with time limits or larger inputs   | Unable to solve within time constraints          |
| <b>Test Case Handling</b>      | 15%       | Handles all test cases including edge and hidden cases                                    | Handles most test cases                                   | Misses important edge cases                   | Fails on multiple test cases                     |

| Q. No. / Criteria | Problem Understanding | Algorithm | Code Implementation | Competitive Performance | Test Case Handling | Sub. Total | Total % |
|-------------------|-----------------------|-----------|---------------------|-------------------------|--------------------|------------|---------|
| 1                 | 15                    | 18        | 15                  | 15                      | 12                 | 75         | 75%     |
| 2                 |                       |           |                     |                         |                    |            |         |
| 3                 |                       |           |                     |                         |                    |            |         |

| Faculty In-charge    | Course Coordinator | Program Director |
|----------------------|--------------------|------------------|
| Dr.T.Kumaragurubaran |                    |                  |
| Signature: _____     | Signature: _____   | Signature: _____ |
| Date: _____          | Date: _____        | Date: _____      |

## CODE: Porter Stemmer

class PorterStemmer:

```

    @staticmethod
    def is_vowel(word, index):
        ch = word[index].lower()

        if ch in "aeiou":
            return True

        if ch == 'y':
            if index == 0:
                return False
            return not PorterStemmer.is_vowel(word, index - 1)

        return False

```

```

    @staticmethod
    def measure(stem):
        m = 0
        prev_vowel = False

        for i in range(len(stem)):
            if PorterStemmer.is_vowel(stem, i):
                prev_vowel = True
            else:
                if prev_vowel:
                    m += 1
                prev_vowel = False

        return m

```

```

    @staticmethod
    def contains_vowel(word):
        for i in range(len(word)):

```

```
    if PorterStemmer.is_vowel(word, i):
        return True
    return False
```

```
@staticmethod
```

```
def ends_with_double_consonant(word):
```

```
    if len(word) < 2:
        return False
```

```
    if word[-1] != word[-2]:
        return False
```

```
    return not PorterStemmer.is_vowel(word, len(word) - 1)
```

```
@staticmethod
```

```
def cvc(word):
```

```
    if len(word) < 3:
        return False
```

```
    i = len(word)
```

```
    if (not PorterStemmer.is_vowel(word, i - 3) and
        PorterStemmer.is_vowel(word, i - 2) and
        not PorterStemmer.is_vowel(word, i - 1)):
```

```
        return word[-1] not in "wxy"
```

```
    return False
```

```
@staticmethod
```

```
def step1a(word):
```

```
    if word.endswith("sses"):
        return word[:-2]
```

```
    if word.endswith("ies"):
        return word[:-2]
```

```
    if word.endswith("ss"):
        return word
```

```
    if word.endswith("s"):
        return word[:-1]
```

```
    return word
```

```
@staticmethod
```

```
def step1b(word):
```

```
    if word.endswith("eed"):
        stem = word[:-3]
```

```
    if PorterStemmer.measure(stem) > 0:
        return stem + "ee"
```

```

        return word

changed = False

if word.endswith("ed"):
    stem = word[:-2]

    if PorterStemmer.contains_vowel(stem):
        word = stem
        changed = True

elif word.endswith("ing"):
    stem = word[:-3]

    if PorterStemmer.contains_vowel(stem):
        word = stem
        changed = True

if changed:

    if word.endswith(("at", "bl", "iz")):
        word += "e"

    elif (PorterStemmer.ends_with_double_consonant(word)
          and word[-1] not in "lsz"):
        word = word[:-1]

    elif (PorterStemmer.measure(word) == 1
          and PorterStemmer.cvc(word)):
        word += "e"

return word

@staticmethod
def step1c(word):

    if word.endswith("y"):
        stem = word[:-1]

        if PorterStemmer.contains_vowel(stem):
            return stem + "i"

    return word

@staticmethod
def step2(word):

    rules = [
        ("ational", "ate"),
        ("tional", "tion"),
        ("enci", "ence"),
        ("anci", "ance"),
        ("izer", "ize"),
        ("abli", "able"),

```

```

("alli", "al"),
("entli", "ent"),
("eli", "e"),
("ousli", "ous"),
("ization", "ize"),
("ation", "ate"),
("ator", "ate"),
("alism", "al"),
("iveness", "ive"),
("fulness", "ful"),
("ousness", "ous"),
("aliti", "al"),
("iviti", "ive"),
("biliti", "ble"),
("logi", "log")
]

```

for old, new in rules:

if word.endswith(old):

stem = word[:-len(old)]

if PorterStemmer.measure(stem) > 0:

return stem + new

return word

return word

@staticmethod

def step3(word):

```

rules = [
("icate", "ic"),
("ative", ""),
("alize", "al"),
("iciti", "ic"),
("ical", "ic"),
("ful", ""),
("ness", "")
]

```

for old, new in rules:

if word.endswith(old):

stem = word[:-len(old)]

if PorterStemmer.measure(stem) > 0:

return stem + new

return word

return word

```

@staticmethod
def step4(word):

    suffixes = [
        "al", "ance", "ence", "er", "ic",
        "able", "ible", "ant", "ement",
        "ment", "ent", "ion", "ou",
        "ism", "ate", "iti", "ous",
        "ive", "ize"
    ]

    for suffix in suffixes:

        if word.endswith(suffix):

            stem = word[:-len(suffix)]

            if suffix == "ion":

                if (PorterStemmer.measure(stem) > 1
                    and len(stem) > 0
                    and stem[-1] in "st"):
                    return stem

            else:
                if PorterStemmer.measure(stem) > 1:
                    return stem

            return word

    return word

@staticmethod
def step5a(word):

    if word.endswith("e"):

        stem = word[:-1]
        m = PorterStemmer.measure(stem)

        if m > 1:
            return stem

        if m == 1 and not PorterStemmer.cvc(stem):
            return stem

    return word

@staticmethod
def step5b(word):

    if (PorterStemmer.measure(word) > 1
        and word.endswith("ll")):
        return word[:-1]

```

```
return word
```

```
@staticmethod  
def stem(word):
```

```
    word = word.lower()
```

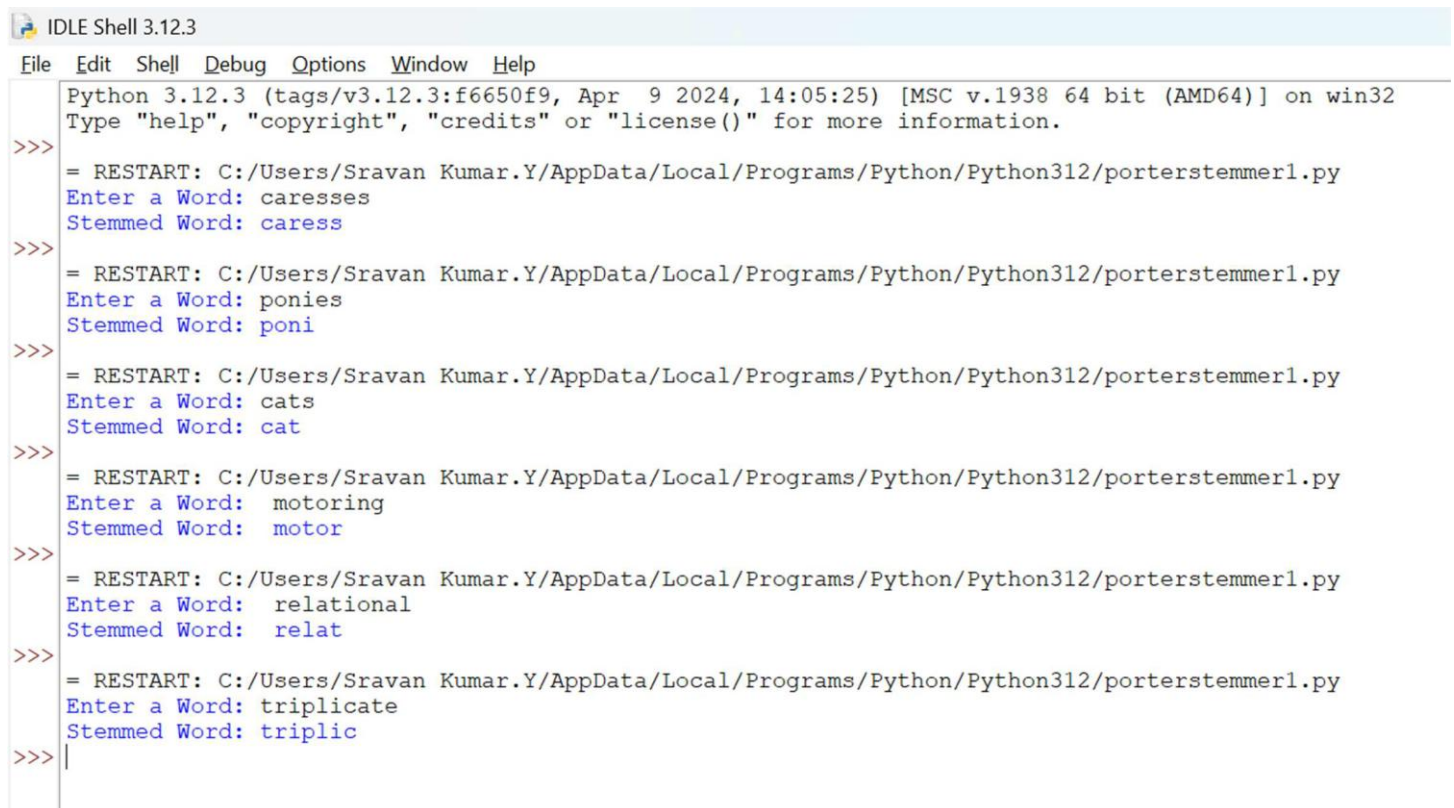
```
    word = PorterStemmer.step1a(word)  
    word = PorterStemmer.step1b(word)  
    word = PorterStemmer.step1c(word)  
    word = PorterStemmer.step2(word)  
    word = PorterStemmer.step3(word)  
    word = PorterStemmer.step4(word)  
    word = PorterStemmer.step5a(word)  
    word = PorterStemmer.step5b(word)
```

```
    return word
```

```
word = input("Enter a Word: ")
```

```
print("Stemmed Word:", PorterStemmer.stem(word))
```

## Output



```
IDLE Shell 3.12.3  
File Edit Shell Debug Options Window Help  
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32  
Type "help", "copyright", "credits" or "license()" for more information.  
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/porterstemmer1.py  
Enter a Word: caresses  
Stemmed Word: caress  
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/porterstemmer1.py  
Enter a Word: ponies  
Stemmed Word: poni  
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/porterstemmer1.py  
Enter a Word: cats  
Stemmed Word: cat  
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/porterstemmer1.py  
Enter a Word: motoring  
Stemmed Word: motor  
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/porterstemmer1.py  
Enter a Word: relational  
Stemmed Word: relat  
>>> = RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/porterstemmer1.py  
Enter a Word: triplicate  
Stemmed Word: triplic  
>>> |
```